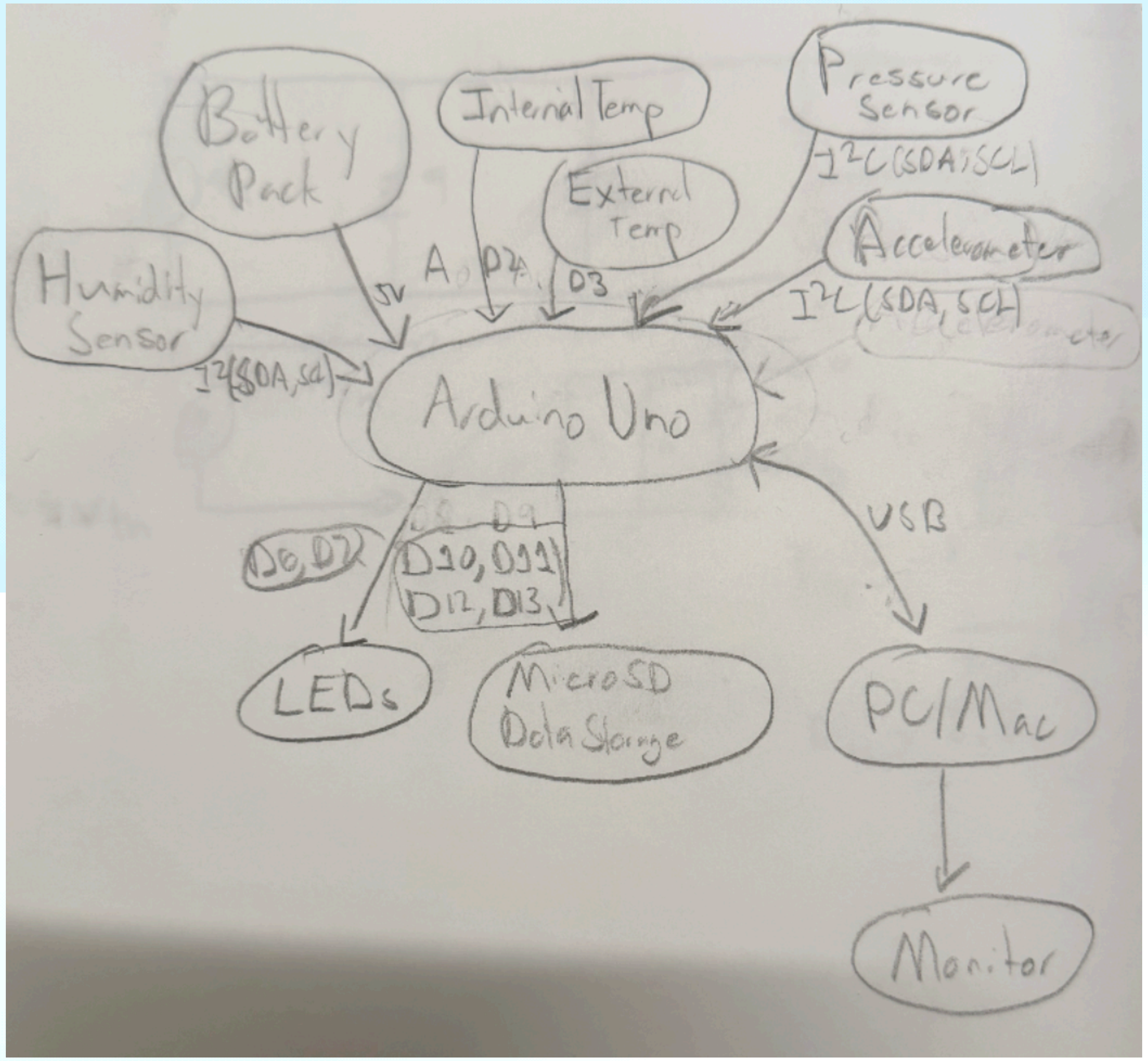
Microprocessors Final Project

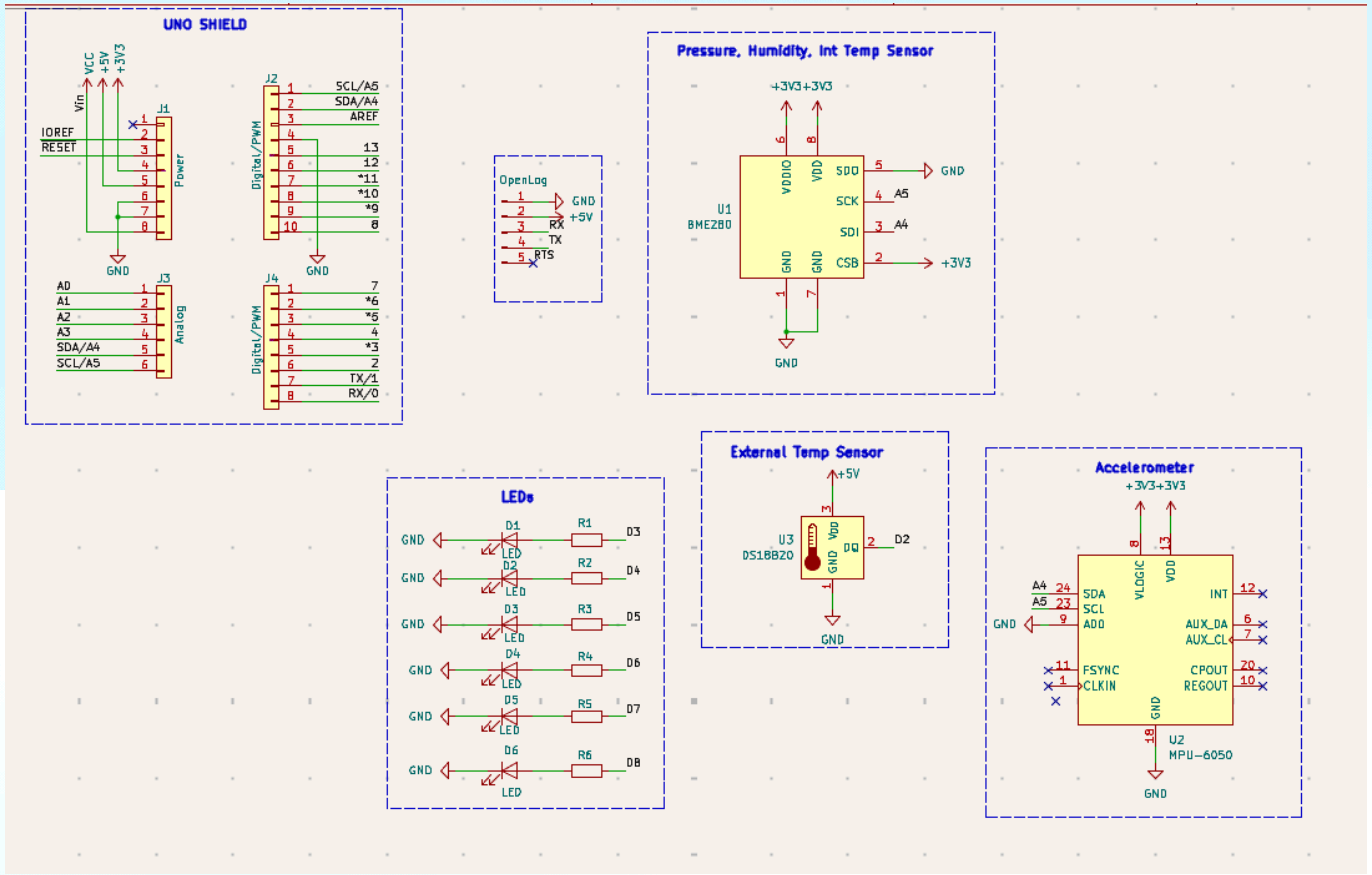
Phase 1



Component	Interface	Estimated Current (mA)	Estimated Power (W @ 5V)	Estimated Weight (g)
Arduino Uno	Main microcontroller	45	0.225	25
Internal Temp Sensor	Environmental sensing	1.5	0.0075	2
External Temp Sensor	Environmental sensing	1.5	0.0075	2
Pressure/Humidity Sensor	Atmospheric sensing	1.0	0.005	3
Accelerometer	Motion sensing	4	0.020	2
MicroSD Module	Data storage	50	0.250	5
LEDs	System status	20	0.100	1
Battery pack	Power source	N/A	N/A	60
Wiring	Connections	N/A	N/A	15

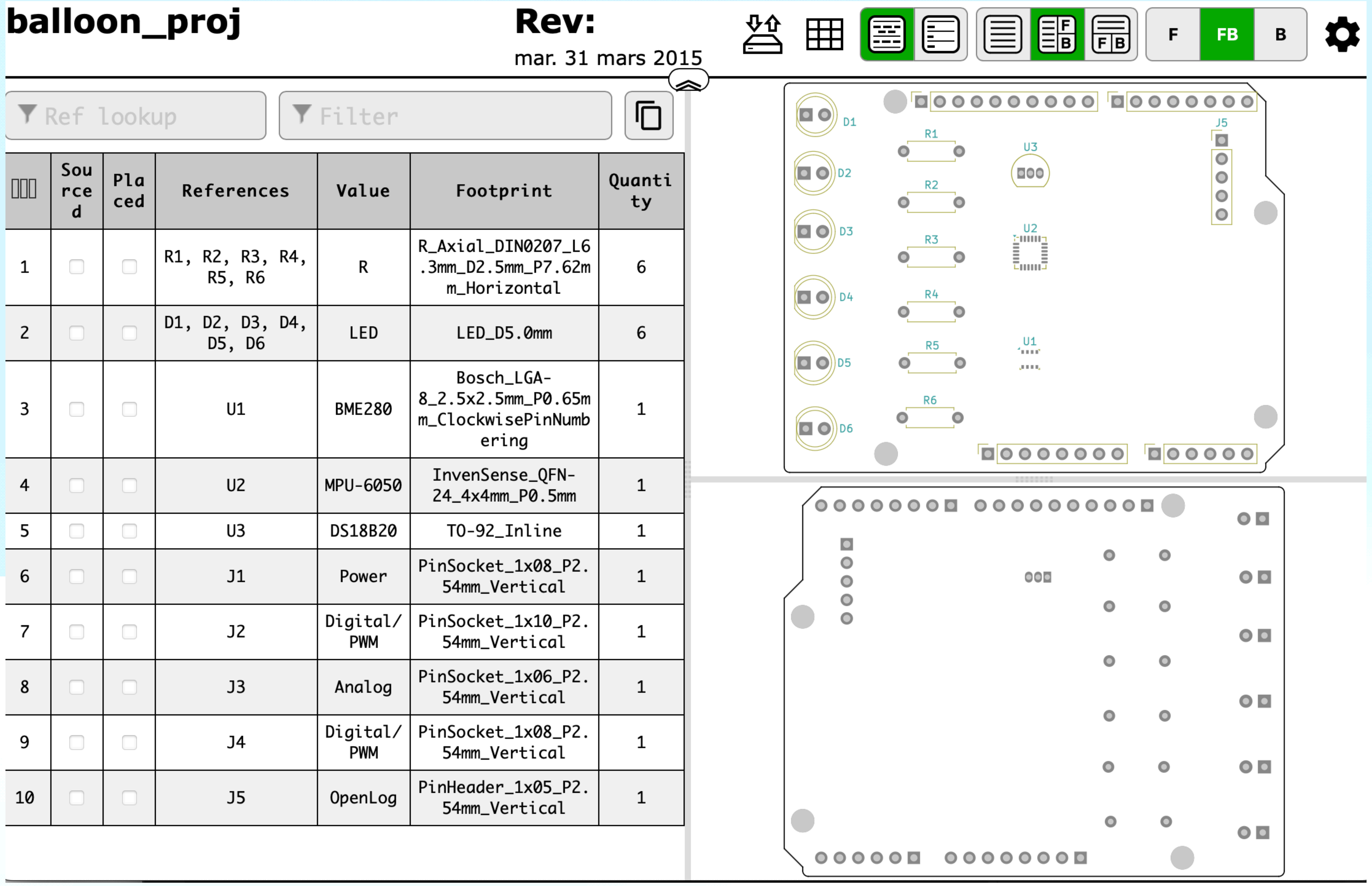
Total Estimated Current = ~130 mA
Total Estimated Power = ~0.62W
Total Estimated Weight = 115g

Phase 2



Component	Qty	Approx Price	Approx Weight	Total Price	Total Weight
Arduino Uno	1	$25	25g	$25	25g
BME280	1	$8	2g	$8	2g
MPU6050	1	$4	2g	$4	2g
DS18B20	1	$3	3g	$3	3g
OpenLog	1	$15	4g	$15	4g
MicroSD card	1	$6	1g	$6	1g
LEDs	1	$0.10	0.3g	$0.60	1.8g
Resistors	1	$0.02	0.1g	$0.12	0.6g
Pin headers	1	$1	3g	$1	3g
PCB	1	$5	20g	$5	20g

Total Cost: ~ $67
Total Weight: ~62g



balloon_proj **Rev:**

mar. 31 mars 2015

	Sourced	Placed	References	Value	Footprint	Quantity
1			R1, R2, R3, R4, R5, R6	R	R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal	6
2			D1, D2, D3, D4, D5, D6	LED	LED_D5.0mm	6
3			U1	BME280	Bosch_LGA-8_2.5x2.5mm_P0.65mm_ClockwisePinNumbering	1
4			U2	MPU-6050	InvenSense_QFN-24_4x4mm_P0.5mm	1
5			U3	DS18B20	TO-92_Inline	1
6			J1	Power	PinSocket_1x08_P2.54mm_Vertical	1
7			J2	Digital/PWM	PinSocket_1x10_P2.54mm_Vertical	1
8			J3	Analog	PinSocket_1x06_P2.54mm_Vertical	1
9			J4	Digital/PWM	PinSocket_1x08_P2.54mm_Vertical	1
10			J5	OpenLog	PinHeader_1x05_P2.54mm_Vertical	1